Primary 3 · Science

Science · Set 1 — Answer Key

80 marks · 1 h 15 min

For parents. Give this to the pupil only after the paper is finished. Written answers: accept any answer with the same meaning, in a complete sentence with correct tense.

Section A: Multiple-Choice Questions (40 marks)

No.	Answer	Marks
1	(3) Plant	2
2	(3) They have wheels	2
3	(1) Fern	2
4	(2) Birds	2
5	(3) Chicken	2
6	(3) Fungi	2
7	(3) Plastic	2
8	(2) A rubber band	2
9	(3) A wooden block	2
10	(1) Seedling	2
11	(3) Cockroach	2
12	(3) Pupa	2
13	(3) Butterfly	2
14	(2) Glass	2
15	(3) Cloud	2
16	(3) Flower	2
17	(2) The length and width of each strip	2
18	(1) The amount of water dripped and the time waited	2
19	(3) Plant C	2
20	(2) Glass	2

Section B: Structured Questions (16 marks)

No.	Answer	Marks
21(a)	True	1
21(b)	False	1
22(a)	True	1
22(b)	False	1
23	Waterproof	2
24	Transparent (clear / see-through)	2
25	(2) The weight or amount of water absorbed	2
26	(2) To make the results more reliable	2
27	Pupa (chrysalis)	2
28(a)	True	1
28(b)	False	1

Section C: Open-Ended Questions (24 marks)

No.	Answer	Marks
29	The plant did not get enough sunlight. Without light, a plant cannot make its own food properly, and its leaves cannot make the green colouring (chlorophyll), so the leaves turned pale and the plant grew weak and thin as it stretched to search for light. — Marking guide: 1 mark for mentioning the lack of sunlight/light, 1 mark for linking this to the plant being unable to make food or green colouring (chlorophyll) without light, 1 mark for describing the effect (pale/yellow leaves, weak or thin growth). Full marks for a clear, complete explanation.	3
30	She could gently touch the snail's body again, perhaps near its head, to see if it responds or pulls back, and she could watch closely for any slow movement, such as breathing near its breathing hole, or place a small piece of food nearby to see if it reacts. — Marking guide: 1 mark for a sensible test of response to touch/stimulus, 1 mark for a second sensible test (e.g. checking for breathing, reaction to food), 1 mark for a clear, complete explanation. Accept other reasonable tests of a sign of life.	3

31	Mushrooms are not plants. They belong to a different group of living things called fungi. Unlike green plants, fungi cannot make their own food using sunlight — they get their food by breaking down dead or living things instead. — Marking guide: 1 mark for stating mushrooms are fungi, not plants; 1 mark for stating fungi cannot make their own food from sunlight (or do not have chlorophyll); 1 mark for a clear overall explanation. Note: saying mushrooms reproduce using spores is true but is NOT the key difference (ferns also use spores) — on its own this earns at most 1 mark, unless the answer also explains that fungi cannot make their own food.	3
32	Thin paper is not strong, so it would tear easily under the weight of heavy books. A better material would be something strong and flexible, such as canvas fabric or thick plastic, which can hold the weight without tearing. — Marking guide: 1 mark for identifying that paper is weak or tears easily, 1 mark for suggesting a suitable alternative material, 1 mark for explaining why the alternative is better (e.g. strong, does not tear).	3
33	A butterfly's life cycle includes a pupa stage, during which its body changes completely from a caterpillar into a butterfly. A cockroach's life cycle has no pupa stage — the nymph already looks like a small adult and simply grows bigger after each moult, without such a big change in body shape. — Marking guide: 1 mark for mentioning the pupa stage in the butterfly's cycle, 1 mark for noting the cockroach has no pupa stage and the nymph resembles the adult, 1 mark for a clear comparison.	3
34	Rubber is a good choice because it is waterproof, so water does not soak through and wet the foot inside. It also bends easily (is flexible) without cracking or breaking, so it is comfortable to walk in and does not slip easily on wet ground. — Marking guide: 1 mark for stating rubber is waterproof, 1 mark for a second correct property (e.g. flexible/bends without breaking, or has good grip and does not slip), 1 mark for a clear overall explanation linking the properties to why rubber suits a rain boot.	3

35	Dropping all three balls from the same height makes the test fair, because the height they are dropped from stays the same for every ball. This means any difference in how high they bounce must be caused by the material of the ball, not by a difference in drop height. — Marking guide: 1 mark for identifying that it keeps the test fair, 1 mark for explaining that only the ball's material should differ, 1 mark for a clear overall explanation. | Explain: Changed: the material of the ball (rubber, glass, foam). Kept the same: the height each ball was dropped from, and the floor. Measured: how high each ball bounced. Dropping height is something that was KEPT THE SAME, not changed or measured.	3
36	A tadpole undergoes a big change in body shape as it grows into an adult frog — it starts with no legs and a tail for swimming, then slowly grows legs and loses its tail. A chick, however, already has a body shape similar to an adult chicken from a young age and mainly just grows bigger as it matures, without such a big change. — Marking guide: 1 mark for describing the tadpole's major body changes, 1 mark for noting the chick already resembles the adult chicken, 1 mark for a clear comparison.	3